

Pramo Jewels

Server Architecture

Document ID: PJ-SAR-001

Version: 1.0

Status: Approved for Deployment

Field	Value
Project	Pramo Jewels Premium E-Commerce Platform
Purpose	Define the production server architecture, infrastructure components and deployment strategy
Audience	DevOps Engineers, System Architects, Developers, Infrastructure Administrators
Version	1.0

Server Architecture Layers

Layer	Components
Edge	DNS, CDN, WAF, SSL/TLS
Load Balancing	Load balancer with health checks
Web Tier	Frontend application servers
Application Tier	Backend API servers and background workers
Data Tier	Database, cache and object storage
Integration Tier	Payment, shipping, email and third-party APIs
Monitoring	Logging, metrics, tracing and alerting
Backup & Recovery	Automated backups and disaster recovery

Objectives

Provide a scalable, highly available and secure server architecture capable of supporting business growth and high traffic.

Topology

Client → CDN/WAF → Load Balancer → Web Servers → Application Servers → Database/Cache/Object Storage with external service integrations.

Scalability

Support horizontal scaling of web and application servers, stateless services and auto-scaling where available.

High Availability

Use redundant servers, health checks, failover mechanisms and replicated data services to minimize downtime.

Security

Enforce network segmentation, firewalls, HTTPS, least-privilege access, secret management and infrastructure audit logging.

Operations

Implement centralized logging, monitoring, automated backups, infrastructure-as-code and configuration management.

Integrations

Integrates with CI/CD, Monitoring, CDN, DNS, Authentication, Payment Gateway, Shipping Providers and Notification Services.

Acceptance Criteria

Infrastructure supports required workloads, remains highly available, passes security validation and is fully documented.